

Study Report



Analyzed File	Bike Frame v8
Version	Autodesk Fusion (2601.1.37)
Creation Date	2025-05-06, 22:26:11
Author	Joyal

☐ Report Properties

Title	Studies
Author	joyal

Simulation Model 1

Study 1 - Static Stress

Study Properties

Study Type	Static Stress
Last Modification Date	2025-05-06, 22:18:38

Settings

General

Contact Tolerance	0.10 mm
Remove Rigid Body Modes	No

Mesh

Average Element Size (% of model size)	
Solids	10
Scale Mesh Size Per Part	Yes
Average Element Size (absolute value)	-
Element Order	Parabolic
Create Curved Mesh Elements	Yes
Max. Turn Angle on Curves (Deg.)	60
Max. Adjacent Mesh Size Ratio	1.6
Max. Aspect Ratio	9
Minimum Element Size (% of average size)	10

Adaptive Mesh Refinement

Number of Refinement Steps	0
Results Convergence Tolerance (%)	20
Portion of Elements to Refine (%)	10
Results for Baseline Accuracy	von Mises Stress

Materials

Component	Material	Safety Factor
BB:1	Aluminum 6061 T6 0 Cold Formed	Yield Strength

Aluminum 6061 T6 0 Cold Formed

Density	2.700E-06 kg / mm ³
Young's Modulus	69000.00 MPa
Poisson's Ratio	0.33
Yield Strength	369.00 MPa
Ultimate Tensile Strength	389.00 MPa
Thermal Conductivity	0.18 W / (mm C)
Thermal Expansion Coefficient	2.360E-05 / C
Specific Heat	896.00 J / (kg C)

Contacts

Mesh

Type	Nodes	Elements
Solids	55449	28392

☐ Load Case1

☐ Constraints

☐ Fixed1

Type	Fixed
Ux	Fixed
Uy	Fixed
Uz	Fixed

☐ Selected Entities

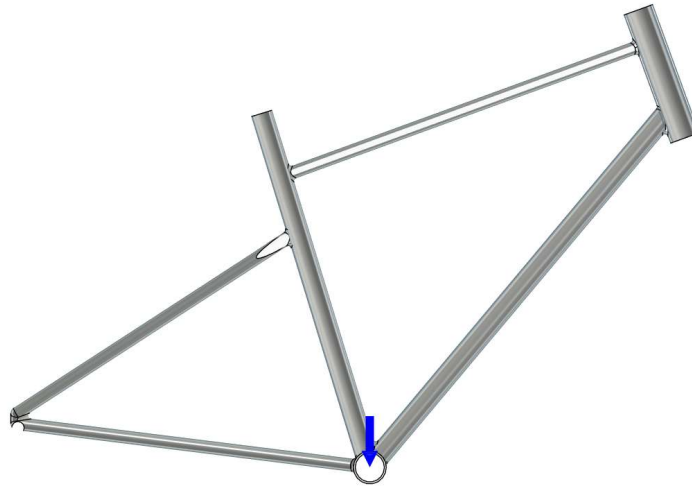


☐ Loads

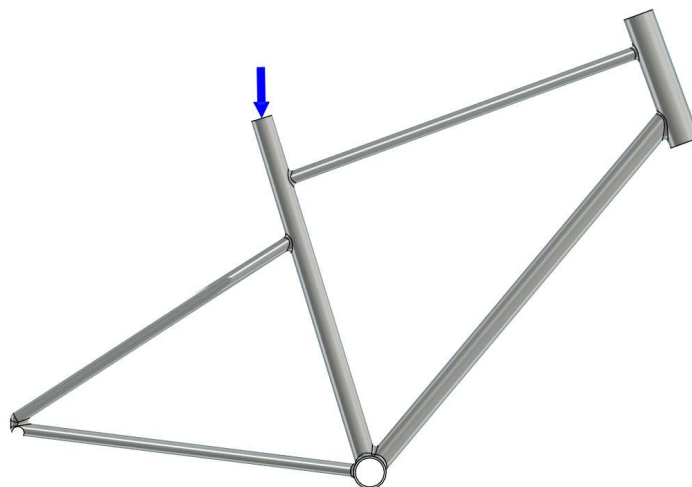
☐ Gravity

Type	Gravity
Magnitude	9.807 m / s ²
X Value	0.00 m / s ²
Y Value	0.00 m / s ²
Z Value	-9.807 m / s ²

☐ Selected Entities

**Force1**

Type	Force
Magnitude	588.60 N
X Value	10.272 N
Y Value	0.00 N
Z Value	-588.51 N
X Angle	-16.0 deg
Y Angle	0.0 deg
Z Angle	0.0 deg
Force Per Entity	No

Selected Entities

Results

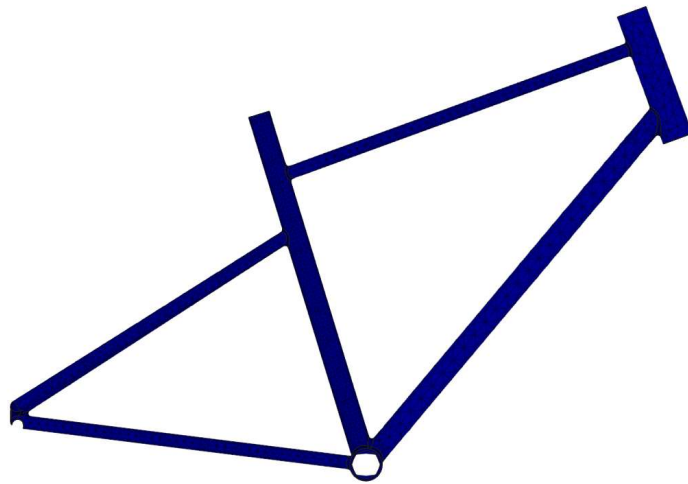
Result Summary

Name	Minimum	Maximum
Safety Factor		
Safety Factor (Per Body)	13.742	15.00
Stress		
von Mises	0.005 MPa	26.852 MPa
1st Principal	-8.567 MPa	30.956 MPa
3rd Principal	-28.30 MPa	7.713 MPa
Normal XX	-11.15 MPa	24.831 MPa
Normal YY	-11.557 MPa	16.549 MPa
Normal ZZ	-28.282 MPa	23.369 MPa
Shear XY	-10.664 MPa	11.55 MPa
Shear YZ	-4.601 MPa	4.514 MPa
Shear ZX	-14.413 MPa	9.026 MPa
Displacement		
Total	0.00 mm	0.278 mm
X	-0.214 mm	0.075 mm
Y	-0.014 mm	0.011 mm
Z	-0.20 mm	0.002 mm
Reaction Force		
Total	0.00 N	294.833 N
X	-128.368 N	186.522 N
Y	-84.975 N	59.313 N
Z	-153.265 N	292.276 N
Strain		
Equivalent	1.283E-07	6.739E-04
1st Principal	8.182E-08	6.601E-04
3rd Principal	-4.926E-04	5.422E-07
Normal XX	-1.500E-04	2.388E-04
Normal YY	-1.959E-04	1.751E-04
Normal ZZ	-3.382E-04	2.360E-04
Shear XY	-4.111E-04	4.453E-04
Shear YZ	-1.774E-04	1.740E-04
Shear ZX	-5.556E-04	3.480E-04
Contact Force		
Total	0.00 N	0.00 N
X	0.00 N	0.00 N
Y	0.00 N	0.00 N
Z	0.00 N	0.00 N

Safety Factor

Safety Factor (Per Body)

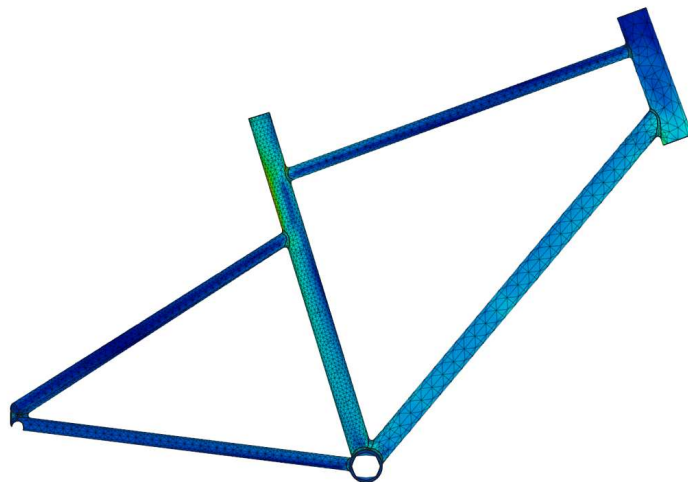
0.00  8.00



Stress

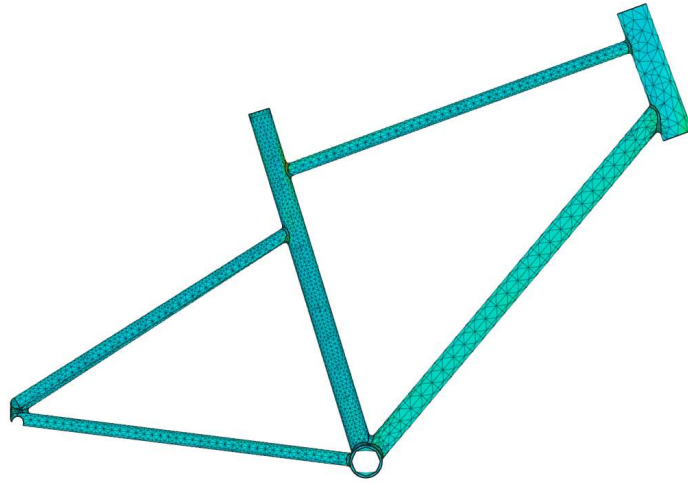
von Mises

[MPa] 0.005  26.852



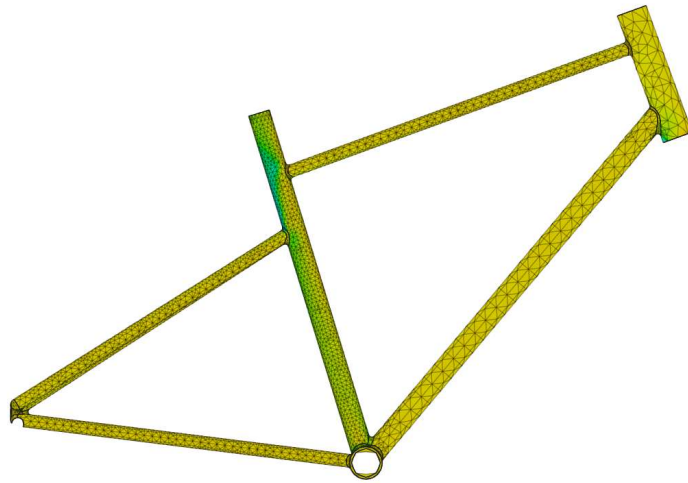
1st Principal

[MPa] -8.567  30.956



3rd Principal

[MPa] -28.30 7.713



Displacement

Total

[mm] 0.00 0.278

